SENTINEL

Privacy-Preserving Autonomous AI Cyber Threat & Fraud Triage Platform

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THEME & NATIONAL TRACK

- **Primary Theme:** Theme 8 — Digital Safety & Cyber Fraud Awareness
- **Secondary Focus:** Theme 7 — AI & Emerging Technologies for Social Good
- **National Vision:** Viksit Bharat 2047 — Digital Sovereignty & Citizen Safety
- **Target UN SDGs:** SDG 9 (Industry, Innovation & Infrastructure) & SDG 16 (Peace, Justice & Strong Institutions)

SOLO INNOVATOR & INSTITUTION CREDENTIALS

- **Sole Innovator / Developer:** Sivabalan T (SEC25CS101 | 3rd Year B.E. CSE)
- **Institution:** Sri Sai Ram Engineering College, Chennai, Tamil Nadu (AISHE: C-16538)
- **Affiliation:** Anna University (Autonomous)
- **Key Milestone:** National Finalist @ Hac'KP 2026 (Kerala Police Cyberdome @ Zoho Corp)

Executive Summary: SENTINEL is an indigenous, open-source-aligned AI SOC Analyst platform that solves enterprise alert fatigue and digital fraud across public infrastructure. It introduces a Zero-Trust In-RAM data sanitizer, a 3-tier hybrid AI routing engine (local Ollama deepseek-r1 to cloud fallback), an AST forensic execution sandbox, and ChromaDB vector threat memory to automate triage in < 30 seconds at 80% reduced cost with zero cloud privacy leakage.

Problem Definition & Beneficiary Analysis

Severe SOC Alert Fatigue, Privacy Breaches & High Licensing Barriers

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CRITICAL CHALLENGES FACED BY INSTITUTIONS

1. **Overwhelming Alert Fatigue:** Enterprise SOCs and university networks receive 5,000+ alerts daily; manual analysis takes 30-45 mins per alert, leaving 70% unexamined.

2. **Citizen Privacy Violations:** Passing raw logs to public cloud LLMs leaks citizen PII, internal IP topologies, and credentials, violating India's DPDP Act 2023.

3. **Cost-Prohibitive Foreign Tools:** Commercial SOAR tools cost millions annually, making them unaffordable for public schools, rural hospitals, and small banks.

WHO IS AFFECTED & THE EXISTING GAPS

• **Primary Beneficiaries:** Public universities, district cooperative banks, healthcare hospitals, municipal smart cities, and law enforcement cyber cells.

• **Existing Gaps in Market:**
- Traditional SIEMs create thousands of false alarms without actionable remediation.
- Cloud AI leaks sovereign Indian data outside national boundaries.
- Air-gapped defense networks have zero access to automated AI intelligence.

Solution Architecture & Core Innovations

Zero-Trust In-RAM Sanitizer, 3-Tier AI Router & Safe Execution Sandbox

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CORE ARCHITECTURAL PILLARS

• **In-RAM Zero-Trust PII Tokenizer:** Replaces 100% of IP addresses, MACs, emails, and credentials with cryptographic tokens strictly in volatile memory.

• **3-Tier Dynamic AI Router:** Routes routine threats to local quantized Ollama (deepseek-r1:8b) at ■0 cloud compute cost; escalates only complex multi-stage attacks.

• **AST-Enforced Forensic Sandbox:** Opcode whitelisting guarantees autonomous scripts run without risk of host escape or unauthorized network calls.

• **ChromaDB Vector Threat Memory:** Semantic cosine similarity matching against 50+ MITRE ATT&CK enterprise playbooks for automated kill-chain correlation.

SYSTEM PIPELINE WORKFLOW

1. Telemetry Ingestion: Wazuh SIEM, Linux Syslog, Suricata JSON webhooks.
↓
2. In-RAM Data Masking: Cryptographic regex/entropy pseudonymizer.
↓
3. 3-Tier AI Routing: Local Ollama → Groq Cloud → Gemini Flash.
↓
4. AST Code Verification: Zero-trust opcode & module inspection.
↓
5. Dual-View RBAC Dashboard: Masked vs Re-identified view with one-click firewall/process containment and instant PDF brief generation.

Technical Stack & Working Prototype

Fully Operational End-to-End Autonomous Incident Response Engine

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TECHNOLOGY STACK SPECIFICATIONS

- **AI & Local Inference:** Ollama, GGUF Q4_K_M Quantization, DeepSeek-R1 (8B/70B), Gemini 2.0 Flash.
- **Vector Database:** ChromaDB with Sentence-Transformers embeddings.
- **Backend Framework:** Python 3.11+ (AsyncIO), FastAPI REST server, WebSockets.
- **Frontend Dashboard:** React 19, Vite, Tailwind UI, Dual-View Analyst Toggle.
- **Reporting Engine:** ReportLab automated executive C-suite PDF compiler.
- **SIEM Ingestion:** Wazuh JSON Webhooks, Syslog, Suricata rule engines.

BENCHMARK VALIDATION & ACCURACY

- **Triage Speed:** Automated alert triage executed in **< 30 seconds** (compared to 30-45 mins manually).

- **Data Privacy Guarantee: 0 bytes** of raw citizen PII or internal IP addresses transmitted outside local host.

- **Cost Efficiency:** Resolves **85-90%** of routine alerts offline at ■0 cloud expense.

- **Proven Recognition:** Selected as **National Finalist at Hac'KP 2026** by Kerala Police Cyberdome @ Zoho Corp Campus.

Impact, Scalability & Viksit Bharat 2047 Alignment

National Security, Digital Sovereignty & Affordable Cyber Defense

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SOCIAL & ECONOMIC IMPACT

• **Digital Sovereignty:** Ensures telemetry data never leaves Indian borders, strictly upholding the Digital Personal Data Protection (DPDP) Act 2023.

• **Democratizing Cyber Safety:** Enables public schools, rural healthcare clinics, and regional banks to operate an autonomous SOC without hiring massive 24/7 analyst teams.

• **Critical Infrastructure Protection:** Shields power grids, smart city networks, and government portals from devastating ransomware.

SCALABILITY & SUSTAINABILITY MODEL

• **Deployment Scalability:** Runs as a lightweight Docker container or edge appliance; scales from a single college server to multi-district state networks.

• **Open-Source Core + Support Model:** Core architecture remains open-source (Apache 2.0) for community adoption, with low-cost support packages for institutional customization.

• **Air-Gapped Defense Readiness:** Operates 100% offline in sensitive law enforcement and military settings with zero internet dependence.

Developer Expertise & Implementation Roadmap

End-to-End Architectural Execution by Sivabalan T (Sri Sai Ram Engg College)

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LEAD ARCHITECT & DEVELOPER

• **Sivabalan T (Sole Innovator & Developer):**
Department of Computer Science & Engineering, Sri Sai Ram Engineering College, Chennai.

• **Core Responsibilities Handled:**
- System Blueprint & Multi-Tier AI Router Architecture
- In-RAM Zero-Trust PII Tokenizer & Privacy Shield
- AST Code Sandboxing & Opcode Whitelisting Engine
- ChromaDB Vector RAG Memory & MITRE ATT&CK Mapping
- Wazuh SIEM Telemetry Webhook Ingestion & Automated PDF Reporting

FOUR-PHASE IMPLEMENTATION ROADMAP

• **Phase 1 (Completed):** In-RAM Sanitizer, 3-Tier AI Router, and AST Sandbox validation.

• **Phase 2 (Completed):** Wazuh SIEM live webhook integration, dual-view RBAC analyst dashboard, and automated PDF reporting.

• **Phase 3 (Next 3 Months):** Pilot deployment in educational campuses and district health centers; local ONNX NER integration.

• **Phase 4 (6-12 Months):** National deployment across state portals and public municipal infrastructure.

AI Tool Disclosure, Open-Source Ethics & Repository

Compliance with IIT Bombay / MoYAS Hack for Social Cause Guidelines

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MANDATORY AI TOOL USAGE DISCLOSURE (PAGE 13-14)

• **Disclosure Statement:** In accordance with Section 2 (Open-Source Commitment) and Section 23 (AI Disclosure), the developer confirms:

• **AI Tools Utilized:** Google Gemini and Antigravity IDE were utilized for initial code scaffolding, boilerplate generation, unit test creation, and technical documentation formatting.

• **Proportion of AI Assistance:** Approximately **25%** of the final implementation involved AI scaffolding, while **75%** (including core security algorithms, in-RAM tokenization logic, AST opcode whitelisting, SIEM ingestion parsers, and system architecture) was independently conceptualized, engineered, and verified by the developer.

OPEN-SOURCE REPOSITORY & VERIFICATION

• **Code Repository:** https://github.com/siva2526k-art/SENTINEL

• **Open-Source License:** Apache License 2.0 (Cleanroom compliant)

• **Documentation Included:** Setup guide, environment configuration, system blueprint, and live demo test scenarios.

• **Self-Certification:** Work is original, free from copyright infringement, and developed by Sivabalan T.